Internship Review

ActionPower
Research Group Intern
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Contents

Vision

- AI Media for Cultural Experiences
  - Preliminary
  - Text -> Image
  - Text + Image -> Image

Speech

- Benchmark Labeling(KOCW)

AI Media for Cultural Experiences

Preliminary

- DPM
- DDPM
- DDIM

Text -> Image

- Stable Diffusion (v1.5, v3)
- Stable Diffusion (LoRA fine-tuned)

Text + Image -> Image

- ControlNet
- IP-Adapter
- IP-Adapter FaceID
- InstantID

AI Media for Cultural Experiences

Goal

Text -> Image : Generate a character image from a text description of a webtoon or fairy-tale character.

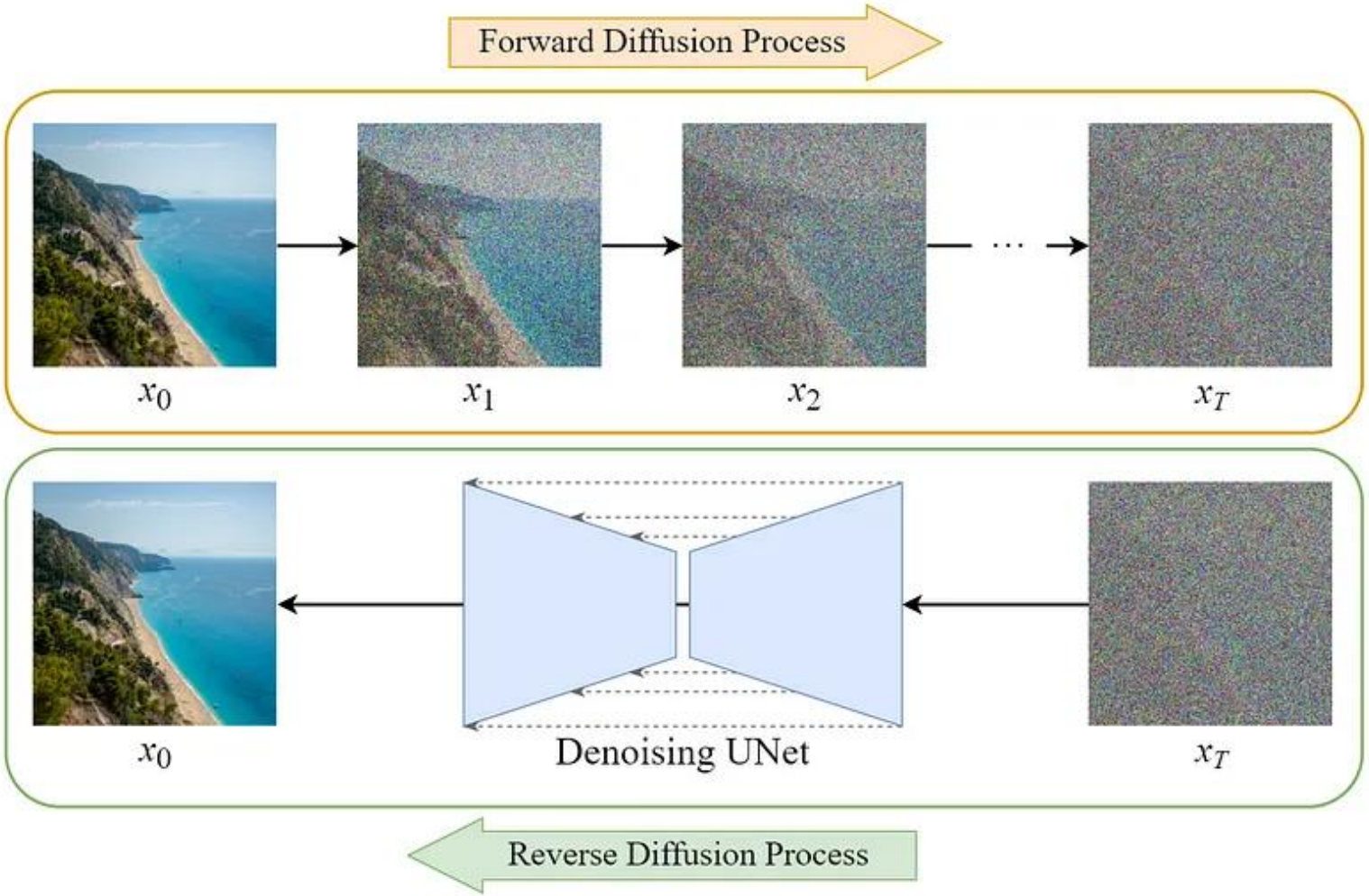
Text + Image -> Image : Given a character image and a prompt describing a scene or action, generate an image of the character performing that action.

Preliminary

DPM (Diffusion Probabilistic Model)

DDPM

DDIM

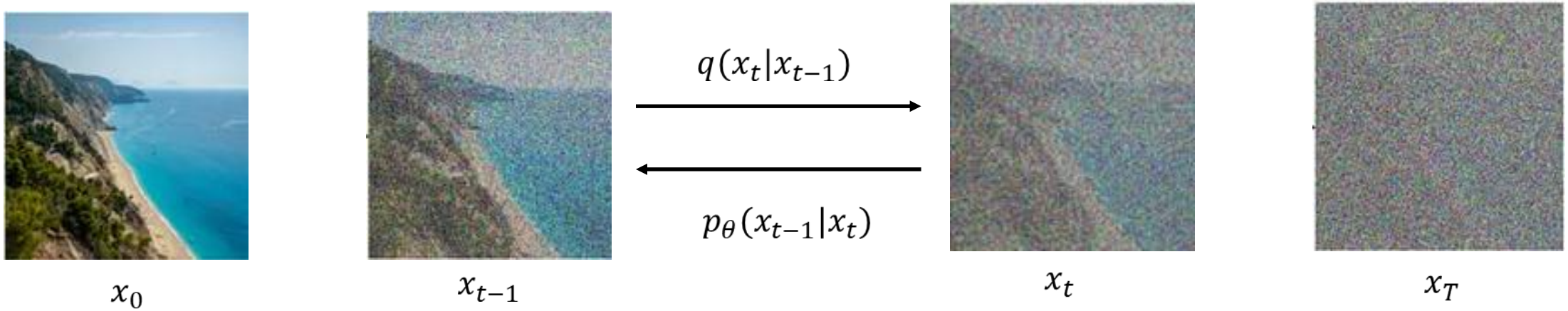


DPM (Diffusion Probabilistic Model)

DDPM

DDIM

1. **Assume Markovian Process**
2. Approximate reverse process to Normal (If $T \rightarrow \infty, q \sim Normal$)



DPM (Diffusion Probabilistic Model)

DDPM

DDIM

Markovian Process

$$q(x_t|x_0, x_1 \ldots, x_{t-1}) = q(x_t|x_{t-1})$$

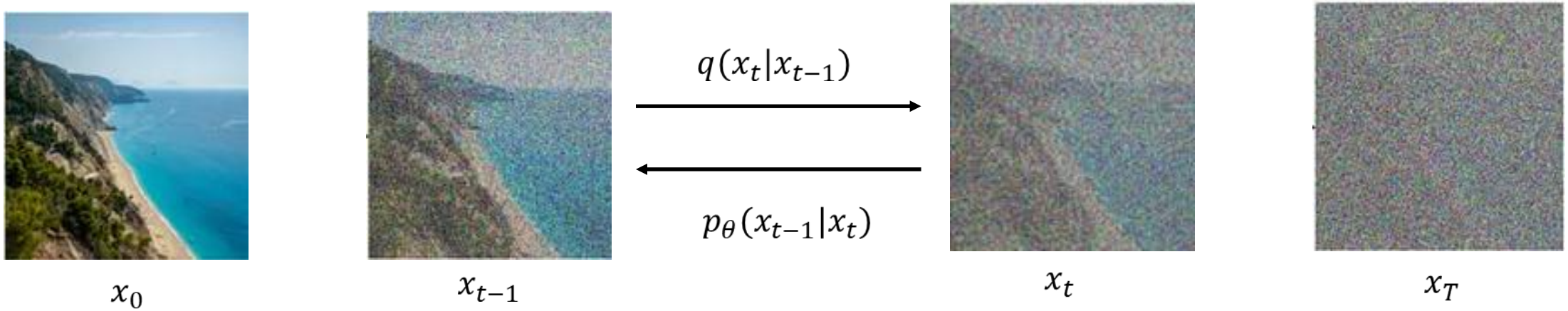
$$p(x_t|x_{t+1}, x_{t+2} \ldots, x_T) = p(x_t|x_{t+1})$$

DPM (Diffusion Probabilistic Model)

DDPM

DDIM

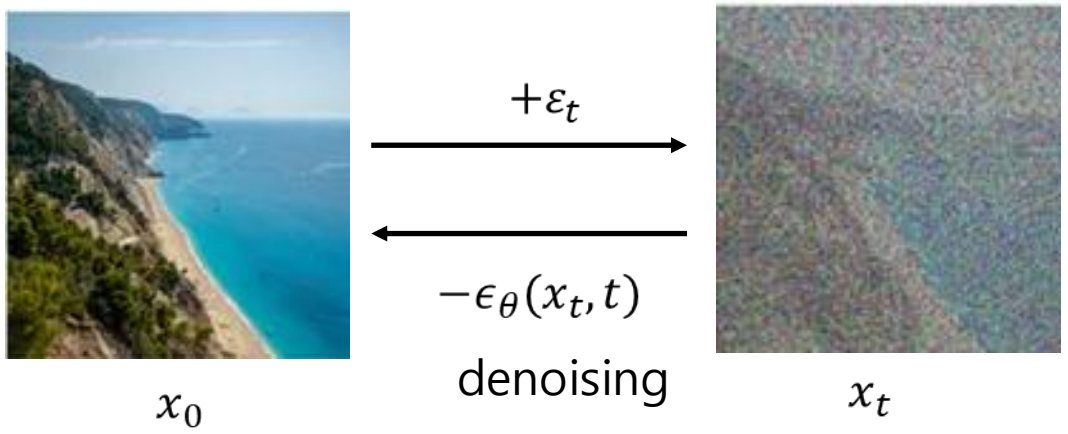
1. Assume Markovian Process
2. **Approximate reverse process to Normal (If $T \rightarrow \infty, q \sim Normal$)**

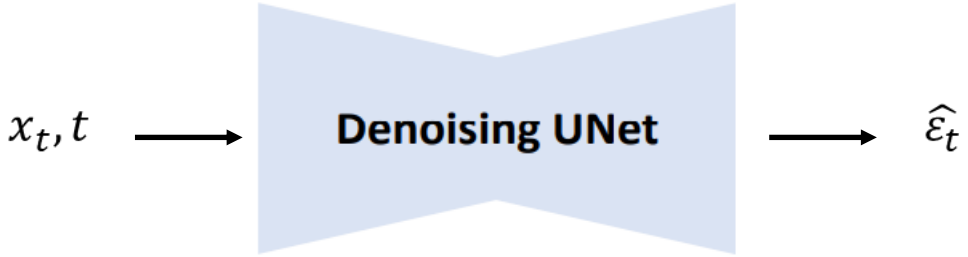


DPM

DDPM(Denoising Diffusion Probabilistic Model)

DDIM



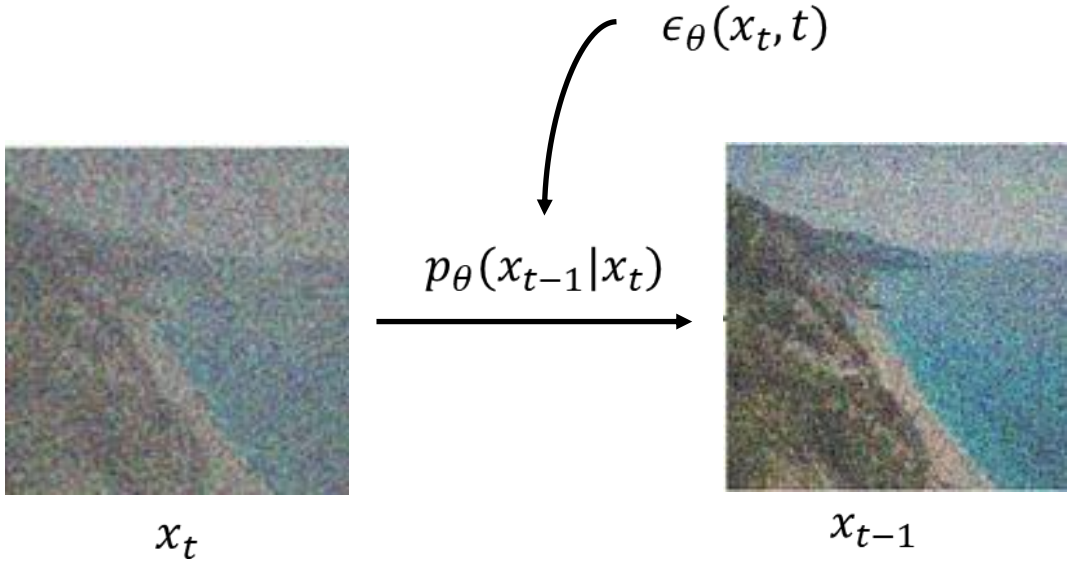


DPM

DDPM(Denoising Diffusion Probabilistic Model)

DDIM

1. **Apply Diffusion Model to real Images**
2. Approximate Loss Function to Simple form



Inference

DPM

DDPM(Denoising Diffusion Probabilistic Model)

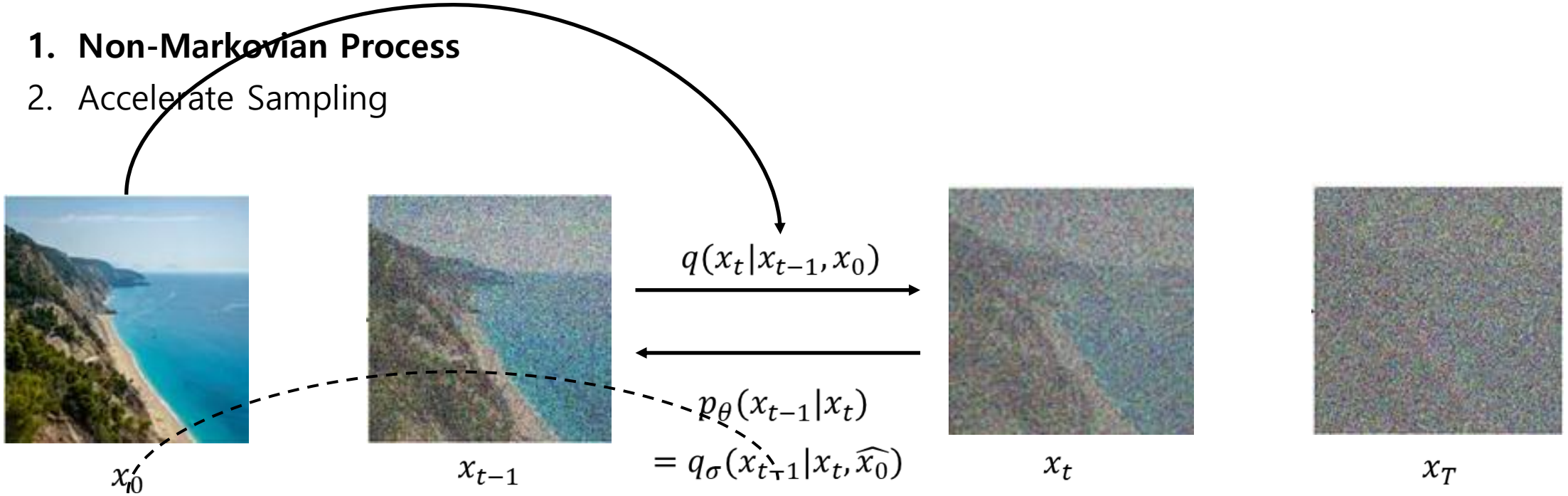
DDIM

1. Apply Diffusion Model to real Images
2. **Approximate Loss Function to Simple form**

$$L_{simple} = E[||\epsilon_t - \epsilon_\theta(x_t, t)||^2]$$

DPM
DDPM
DDIM(Denoising Diffusion Implicit Model)

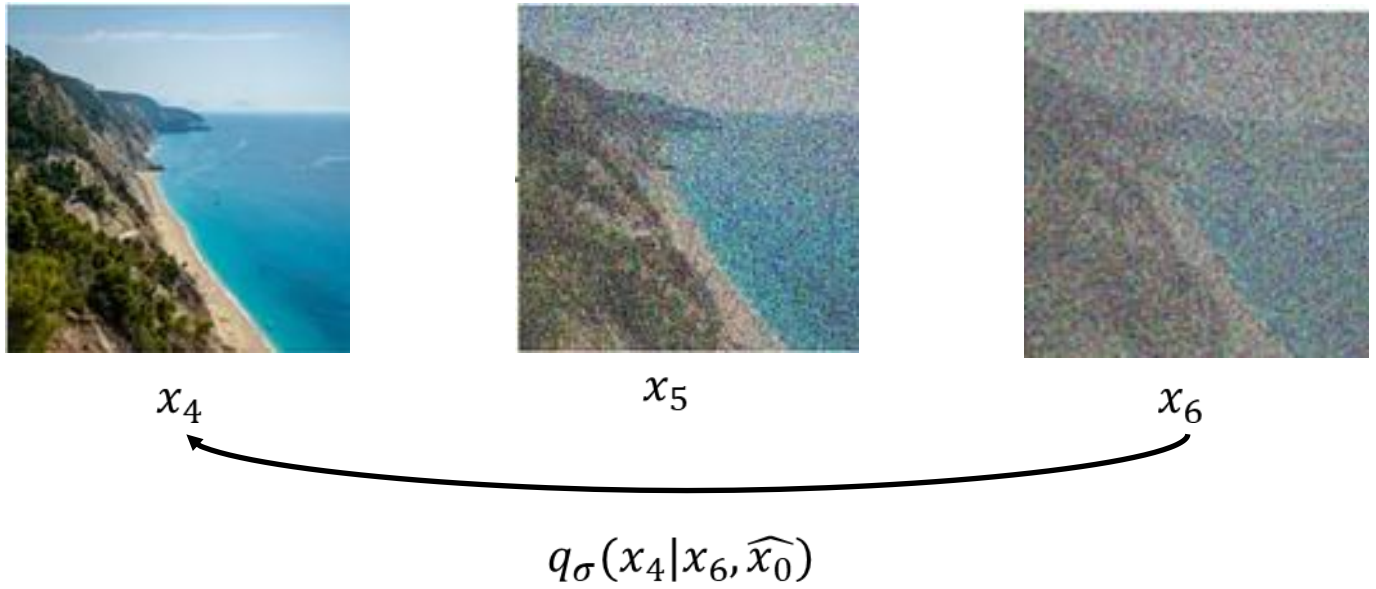
1. **Non-Markovian Process**
2. Accelerate Sampling



Non- Markovian Process

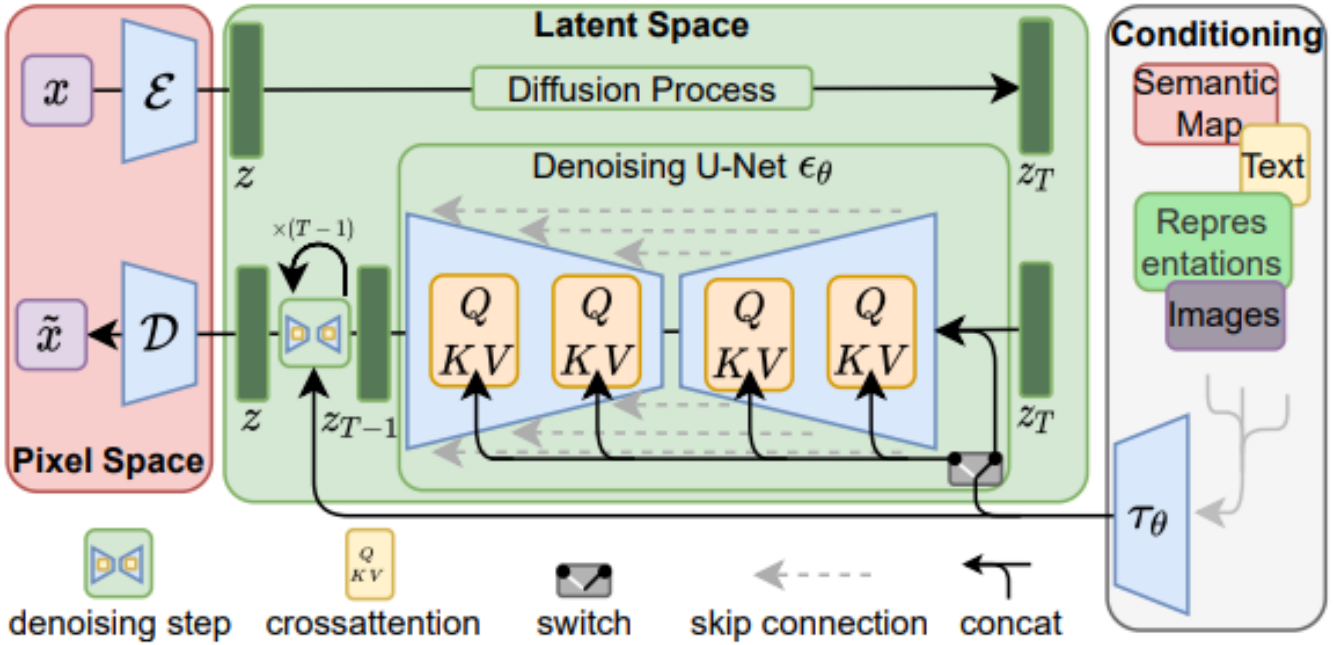
DPM
DDPM
DDIM(Denoising Diffusion Implicit Model)

1. Non-Markovian Process
2. **Accelerate Sampling**



Text -> Image

Stable Diffusion



VAE

Text Encoder

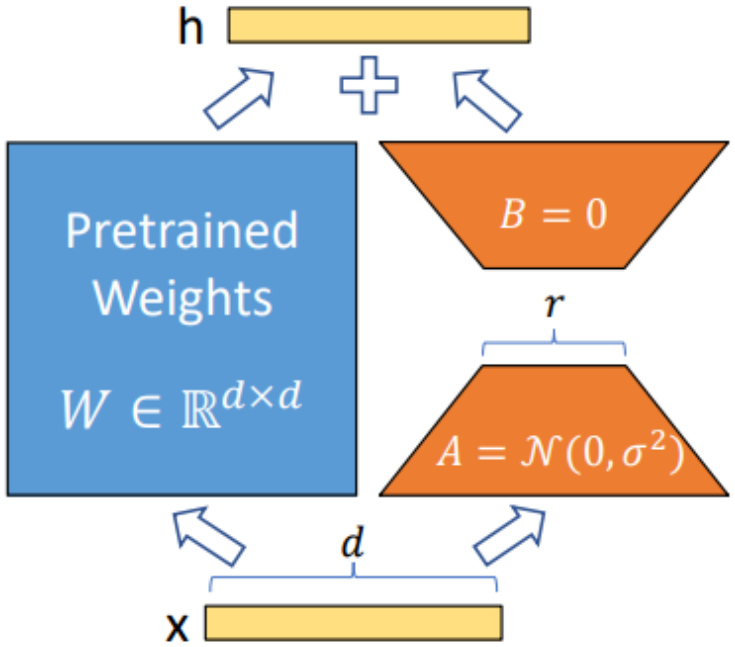
CLIP TextModel

Denoising Unet

Scheduler

DDPM, DDIM, ...

LoRA



$$h = W_0 x + \Delta W x = W_0 x + BAx$$

Train

Stable Diffusion v1.5
Stable Diffusion v3
Fine-Tuned



European



Arab



Asian

Prompt = "Face focus, face of handsome man, blonde short hair, blue eyes, big nose, **[....]** looks, little smile at viewer."

Neg_Prompt = "ugly, tiling, poorly drawn hands, poorly drawn feet, poorly drawn face, out of frame, extra limbs, disfigured, deformed, body out of frame, bad anatomy, watermark, signature, cut off, low contrast, underexposed, overexposed, bad art, beginner, amateur, distorted face"

Stable Diffusion v1.5
Stable Diffusion v3
Fine-Tuned





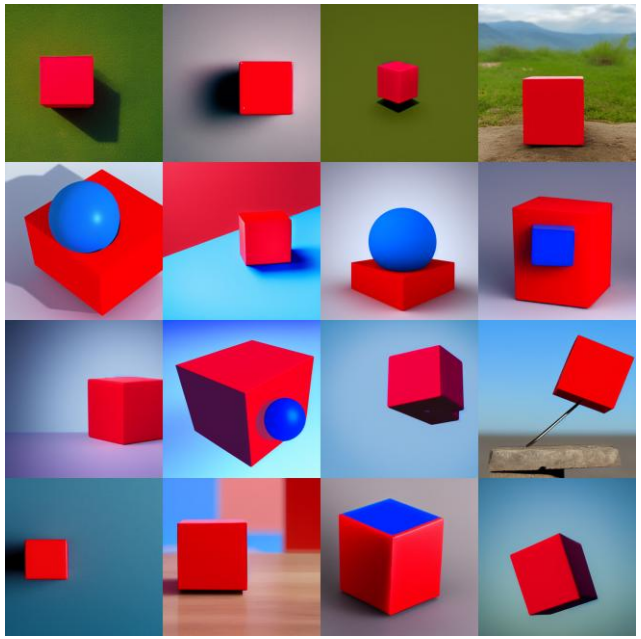
Prompt = "cat, airbrush style, Last winter days, cute kawaii cat, snowy, soft lighting, detailed face, concept art, digital painting, epic, 32k, intricate details, sharp focus, trending on artstation., kids story book style, muted colors, watercolor style."

Neg_Prompt = "2 heads, elongated body, 2 faces, cropped image, out of frame, draft, deformed hands, signature, watermark, twisted fingers, double image, long neck, extra limb, ugly, poorly drawn hands, missing limb, ugly, over saturated, grain, blurry, bad anatomy, disfigured, poorly drawn face, mutation, disconnected limbs"

Stable Diffusion v1.5
Stable Diffusion v3
Fine-Tuned

Limitation

A red cube on top of a blue sphere



Not well trained on recent webtoons



Stable Diffusion v1.5
Stable Diffusion v3
Fine-Tuned

Trained on 1B images
+
Fine-Tuned with 30M high-quality aesthetic images

Good for photorealistic image



Stable Diffusion v1.5
Stable Diffusion v3
Fine-Tuned



European



Arab



Asian

Prompt = "Face focus, face of handsome man, blonde short hair, blue eyes, big nose, **[....]** looks, little smile at viewer."

Neg_Prompt = "ugly, tiling, poorly drawn hands, poorly drawn feet, poorly drawn face, out of frame, extra limbs, disfigured, deformed, body out of frame, bad anatomy, watermark, signature, cut off, low contrast, underexposed, overexposed, bad art, beginner, amateur, distorted face"

Stable Diffusion v1.5
Stable Diffusion v3
Fine-Tuned



Prompt = "A masterpiece photo of a beautiful boy, korean webtoon style, highschool student"

Neg_Prompt = "worst quality, five fingers"



Prompt = "Portrait of cute infantile fairytale 1boy, high detailed, light brown short hair, blue eyes, snub nose, looking_into_camera, watercolor style, pixiv, artstation"

Neg_Prompt = "ugly, tiling, poorly drawn hands, poorly drawn feet, poorly drawn face, out of frame, extra limbs, disfigured, deformed, body out of frame, bad anatomy, watermark, signature, cut off, low contrast, underexposed, overexposed, bad art, beginner, amateur, distorted face"

Stable Diffusion v1.5
Stable Diffusion v3
Fine-Tuned

Mann-E Dreams(XL based)

Train Midjoureny generated images



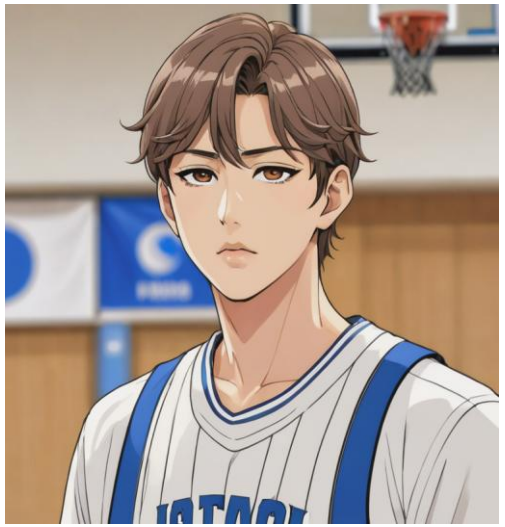
Dreamshaper8(1.5 based)

Good for realistic portraits, anime-style



Stable Diffusion v1.5
Stable Diffusion v3
Fine-Tuned

Mann-E Dreams(XL based)
Train Midjoureny generated images




Dreamshaper8(1.5 based)
Good for realistic portraits, anime-style




LoRA: Satoru Gojo / Jujutsu Kaisen















Prompt = A masterpiece drawing of gojou satoru, at basketball court, sunglasses, round eyewear, highschool basketball player, white and blue striped uniform, dribbling <lora:gojosatoruxl1:0.7> <lora:add-detail-xl:0.5>

Neg_Prompt = photorealistic, ugly, tiling, poorly drawn hands, poorly drawn feet, poorly drawn face, out of frame, extra limbs, disfigured, deformed, body out of frame, bad anatomy, cut off, underexposed, overexposed, bad art, beginner, amateur, distorted face, detailed eye







Prompt = A masterpiece drawing of gojou satoru, at basketball court, sunglasses, round eyewear, highschool basketball player, white and blue striped uniform, dunk shooting , jumping <lora:gojosatoruxl1:0.7> <lora:add-detail-xl:0.2>

Neg_Prompt = photorealistic, ugly, tiling, poorly drawn hands, poorly drawn feet, poorly drawn face, out of frame, extra limbs, disfigured, deformed, body out of frame, bad anatomy, cut off, underexposed, overexposed, bad art, beginner, amateur, distorted face, detailed eye

Webtoon Text → Image

Adding a LoRA to a model fine-tuned for webtoon images produced the best results.

Ex) Mann-E Dreams + gojo satoru LoRA + Detail LoRA

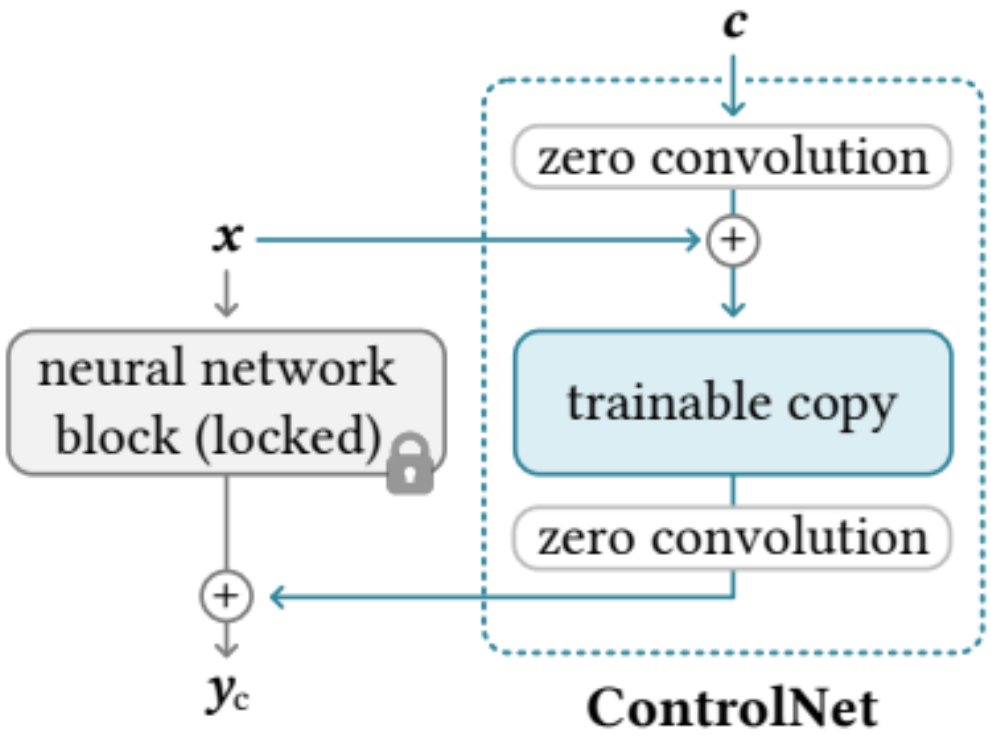
Adding a LoRA to a standard base model produced lower-quality results.

Ex) Stable Diffusion v1.5+ gojo satoru LoRA + Detail LoRA

Text + Image -> Image

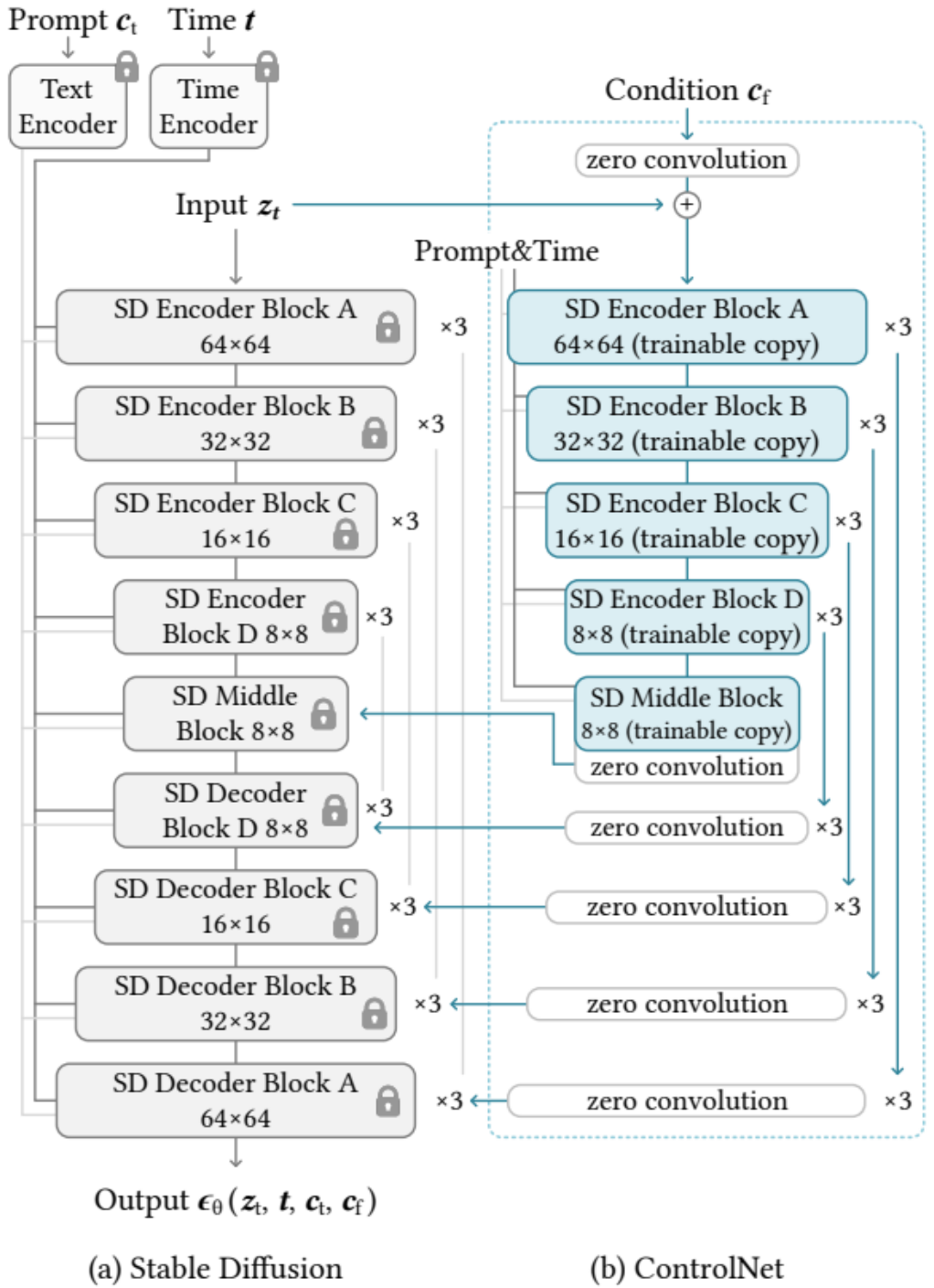
ControlNet
IP-Adapter
InstantID

Train copied NN, zero-initialized convolution



ControlNet
IP-Adapter
InstantID

Limitation : Large Size

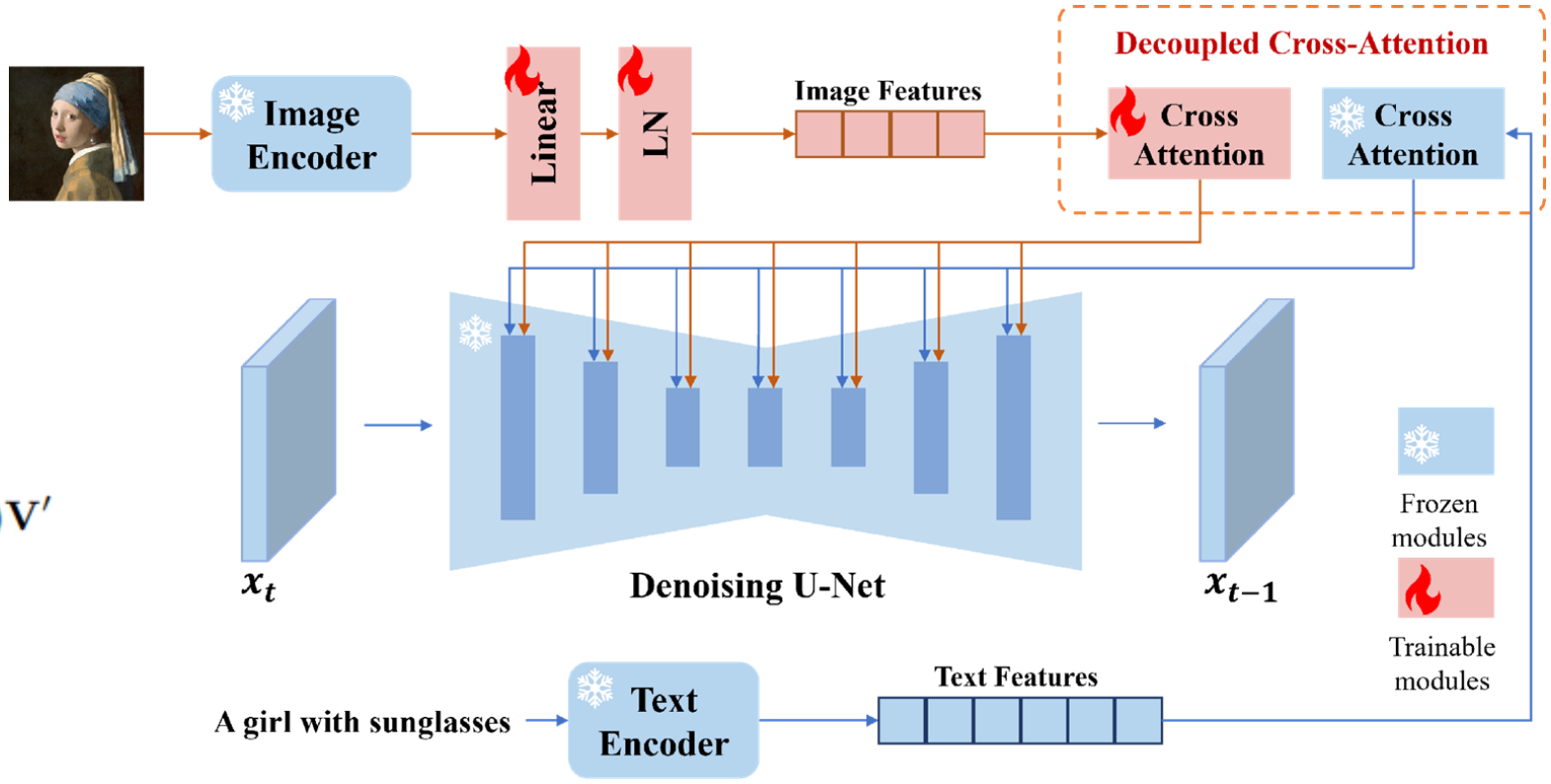


ControlNet
IP-Adapter
InstantID

Feed text and image information into each U-Net cross-attention layer

Lightweight

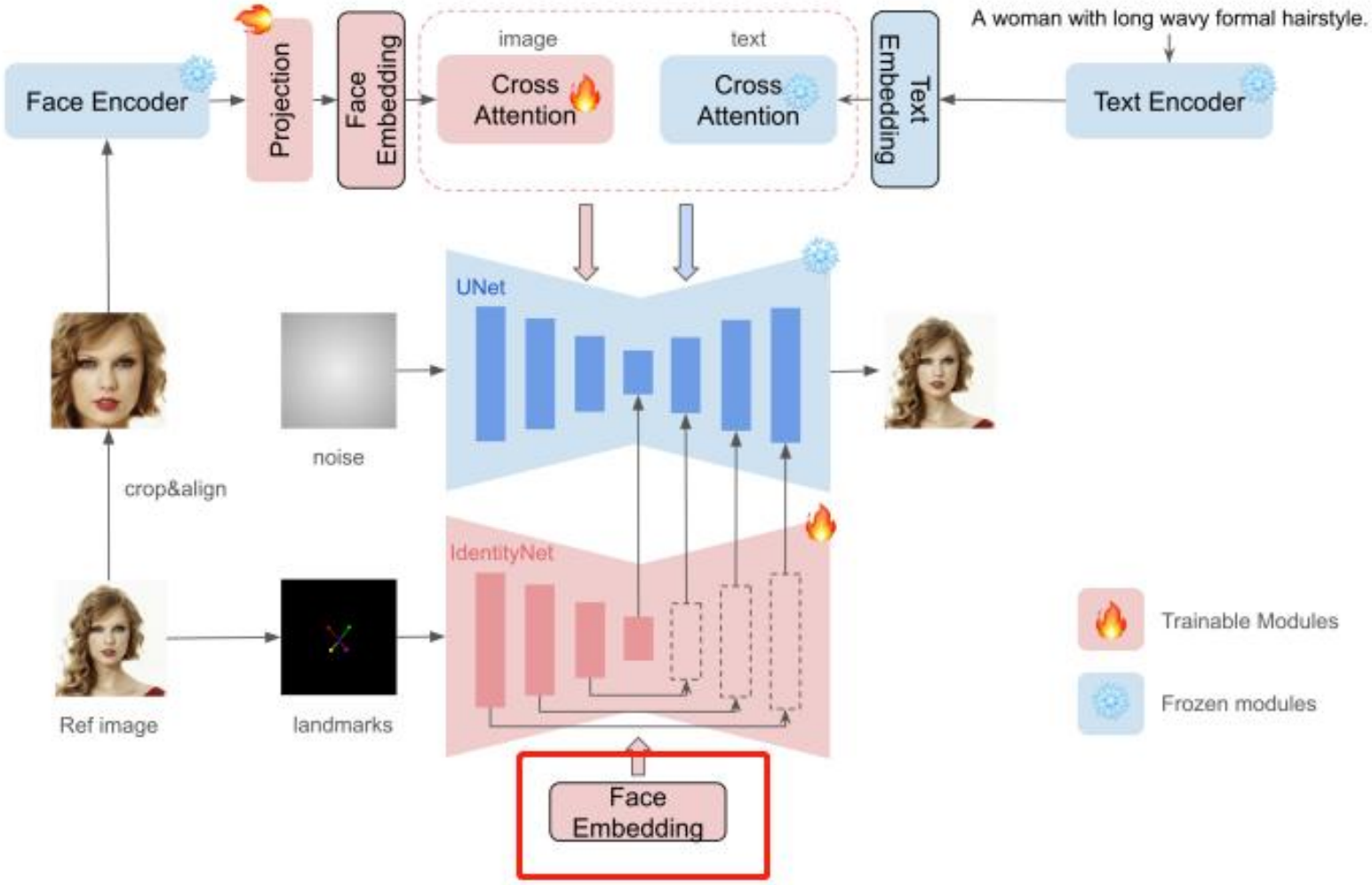
$$\mathbf{Z}^{new} = \text{Softmax}(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}})\mathbf{V} + \text{Softmax}(\frac{\mathbf{Q}(\mathbf{K}')^\top}{\sqrt{d}})\mathbf{V}'$$



ControlNet
IP-Adapter
InstantID

Inputs are mostly portraits

IdentityNet preserves facial identity



ControlNet
IP-Adapter
InstantID





- diffusion model : v1-5-pruned.ckpt
- Positive : handsome athlete boy, webtoon style + best quality
- Negative : lowres, bad anatomy, bad hands, cropped, worst quality
- Control Strength : 1.00, 0.43 for each



- diffusion model : anything-v3-full.safetensors
- Positive : handsome athlete boy(+3d style) + masterpiece, best quality, ultra-detailed, illustration, disheveled hair
- Negative : longbody, lowres, bad anatomy, bad hands, missing fingers, pubic hair,extra digit, fewer digits, cropped, worst quality, low quality
- Control Strength : 1.00 for both

ControlNet
IP-Adapter
InstantID



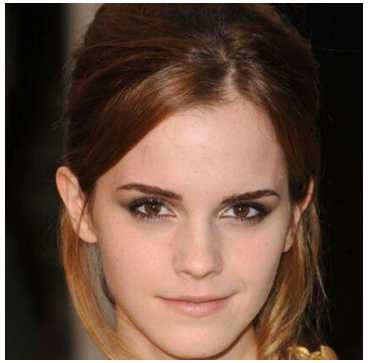
- diffusion model : 1.5 based (dreamlike-art/dreamlike-anime-1.0)
- Positive : photo of a handsome boy wearing highschool uniform(,smiling)
- Negative :
- 50steps





- diffusion model : sdxl based (mann-e/Mann-E_Dreams)
- Positive :best quality, high quality, (gathering with friends), wearing highschool uniform, (cold and aloof demeanor), handsome, (tall)
- Negative : (flushing)
- 50steps

ControlNet
IP-Adapter-FaceID
InstantID







Base Model-
SG161222/Realistic_Vision_V4.0_noVAE

prompt = "photo of a woman wizard"

negative_prompt = "monochrome, lowres, bad anatomy, worst quality, low quality, blurry"

num_inference_steps=30, guidance_scale=7.5

ControlNet
IP-Adapter-FaceID
InstantID



Base Model : Mann-E/Dreams

prompt = "playing basketball, webtoon style"

negative_prompt = "monochrome, lowres, bad anatomy, worst quality, low quality, blurry"

num_inference_steps=30, guidance_scale=7.5





ControlNet
IP-Adapter
InstantID








```
 base_model_path = "mann-e/Mann-E_Dreams"

    prompt = "anime artwork {highschool basektball player }. webtoon style, key visual, vibrant, studio anime,  highly detailed"

    n_prompt = "photo, deformed, black and white, realism, disfigured, low contrast"

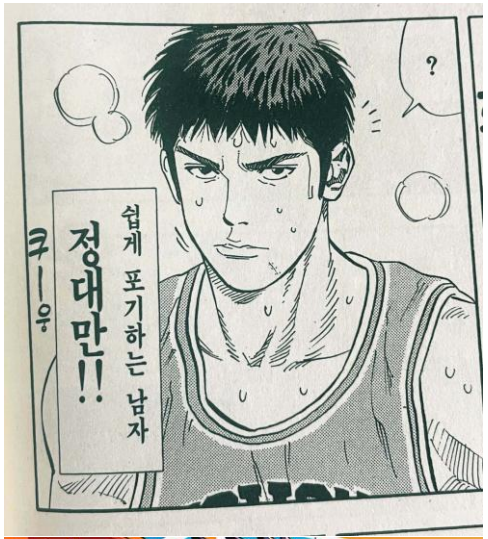

    image = pipe(
        prompt=prompt,
        negative_prompt=n_prompt,
        image_embeds=face_emb,
        image=face_kps,

controlnet_conditioning_scale=0.6,(0.8)
        ip_adapter_scale=0.8,
        num_inference_steps=30,
        guidance_scale=5,
                    ).images[0]
```

ControlNet
IP-Adapter
InstantID














```
base_model_path = "mann-e/Mann-E_Dreams"
    prompt = "anime artwork {basektball player (laying up)(shooting)(playing basketball)}. webtoon style, key visual, vibrant, studio anime, highly detailed, (red short haired)"
    n_prompt = "photo, deformed, black and white, realism, disfigured, low contrast"

    image = pipe(
        prompt=prompt,
        negative_prompt=n_prompt,
        image_embeds=face_emb,
        image=face_kps,
        controlnet_conditioning_scale=0.6,(0.8)
        ip_adapter_scale=0.8,
        num_inference_steps=30,
        guidance_scale=5,
                    ).images[0]
```

Webtoon Text + Image → Image

Choosing the right base model is important for IP-Adapter-FaceID.

InstantID fixes the face position, so the input image needs adjustment beforehand.

Labeling

Labeling

KOCW

Labeling university lecture videos

Engineering (Chemical): Material Balances I, Lecture 3

Engineering (Architecture): History of Western Architecture, Lecture 10

Engineering (Transportation): Railway History and Development, Lecture 5

Engineering (Mechanical/Metals): Building Services, Lecture 4

Engineering (Other): Future Mobility, Lecture 9-1

Engineering (Industrial): The Fourth Industrial Revolution and Modern Technology, Lecture 5

Engineering (Bio/Chem/Env): General Chemistry, Lecture 6

Engineering (Electrical/Electronics): Semiconductor Nano Devices, Lecture 8

Engineering (Computing/Communications): Software Engineering, Lecture 7-1

Arts & Sports (Dance/PE): Dance Aesthetics, Lecture 10

Arts & Sports (Theater/Film): Performing Arts, Lecture 10

Arts & Sports (Music): Introduction to Contemporary Music, Lecture 6

Arts & Sports (Applied Arts): Convergence Art in the Digital Age, Lecture 1-2

Arts & Sports (Dance/PE): Understanding Sports, Lecture 2

Labeling

Difficulties

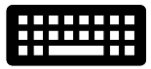
- **Equations, subscripts, and mixed English**
- Abbreviations
- Dialects and pronunciation
- Terms that require subject knowledge



"N A three times ten to the minus sixteen"



$N_a \times 3 \times 10^{16}$



"N_a, three times ten to the sixteenth"

Labeling

Difficulties

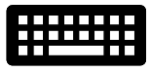
- **Equations, subscripts, and mixed English**
- Abbreviations
- Dialects and pronunciation
- Terms that require subject knowledge



"N A, let's use it. Right, N A Q N A W D max"



N_a쓰잖아 그렇지 N_aqW_{max}



"Na, you know. Right, NaqWmax"

Labeling

Difficulties

Equations, subscripts, and mixed English

Abbreviations

Dialects and pronunciation

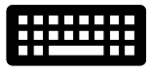
Terms that require subject knowledge



"C ox over Q S S prime"



$$\frac{Q_{SS'}}{C_{ox}}$$



"QSS' divided by Cox"

Labeling

Difficulties

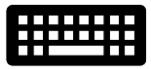
- **Equations, subscripts, and mixed English**
- Abbreviations
- Dialects and pronunciation
- Terms that require subject knowledge



"ee-kway-jun, ee-kkol"



equation, equal



"equation, equal"

Labeling

Difficulties

Equations, subscripts, and mixed English

Abbreviations

Dialects and pronunciation

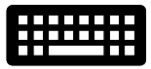
Terms that require subject knowledge



"geun-sa-hwan"



"the Dept. of Civil & Environmental Engineering"



"Geonsahwan" (department abbreviation)

Labeling

Difficulties

Equations, subscripts, and mixed English

Abbreviations

Dialects and pronunciation

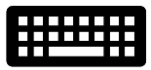
Terms that require subject knowledge



"a lifetime plane, a coal plane, the army"



"a plane shaped like this, a passenger plane, a place like that"



"a plane shaped like this, a passenger plane, a place like that"

Labeling

Difficulties

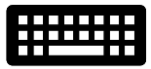
- Equations, subscripts, and mixed English
- Abbreviations
- Dialects and pronunciation
- **Terms that require subject knowledge**



"Pictures at an Exhibition, The Soldier's Tale"



"Mussorgsky: Pictures at an Exhibition, Stravinsky: The Soldier's Tale"



"Pictures at an Exhibition, The Soldier's Tale"

Labeling

Time

Arts & sports:	mean 3.5 h	range 2.5–5.0 h
Engineering:	mean 4.5 h	range 3.0–5.0 h

Labeling one hour of lecture video takes 4–4.5 hours on average.
One hour of work covers 12–15 minutes of lecture video.

Labeling

Task - Time

Timestamps 6

Typos / grammar 3

Technical terms 1

Labeling

Labeling Guideline

How should we label equations, subscripts, and Konglish?

e.g., phonetic “ee-kway-shun” (equation), “ee-kkol” (equal)

A consistent guideline would help across different videos.

Labelers with some background in the lecture topic would be better suited to the task.

Thank you

To Jaesung, for mentoring me throughout my internship and sharing thoughtful feedback
To Hwanbok, Jaejin, and Haeju, for answering my questions and looking after me at mealtimes
And to everyone at ActionPower, for welcoming me so warmly